

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2013

Subject: 340152 Electrical Measuring Instruments

Section: 5 - 6

Date: 22 December 2013

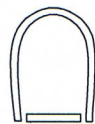
Time: 13.00 - 15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in all classes registered for the current semester.
Student who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. This will be a close-book test.
4. Calculation devices are allowed.
5. Electronic communication devices are NOT allowed in the examination room.
6. The examination has 7 pages (including this pages), 12 questions and total score of 70 marks

Q1) Identify each of the instrument symbols illustrated in Figures. (5 marks)



a)



b)

Class 2.5



c)

d)



e)

Q2) Define the following terms:

(5 marks)

(a) Error :

(b) Accuracy :

(c) Precision :

(d) Measurement :

(e) Instrument :

Q3) List three advantages that electronic instruments have over electronic instruments.

(3 marks)

.....

Q4) Explain Gross, Systematic, and Random Errors.

(3 marks)

.....

Q5) List and explain the three forces involving in the moving system of a deflection instrument.

(3 marks)

.....

Q6) Explain the characteristic of PMMC scale (linear or nonlinear)

(3 marks)

.....

Q7) Sketch the basic construction of a typical PMMC instrument.

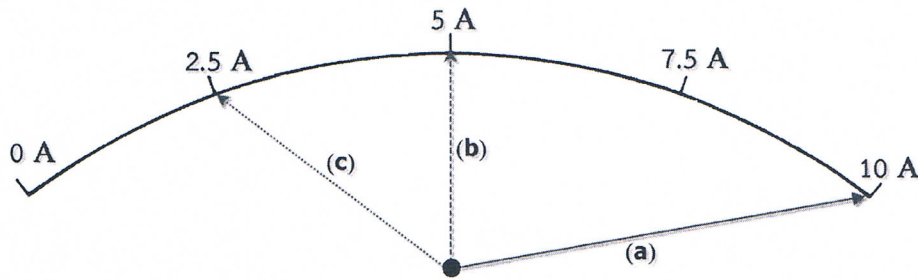
(3 marks)

[illegible]

Q8) The expected value of a resistor is $10\text{ k}\Omega$, but the measurement using ohmmeter yields a value of $9.95\text{ k}\Omega$. Calculate (a) absolute error, (b) % error, and (c) % of accuracy. (5 marks)

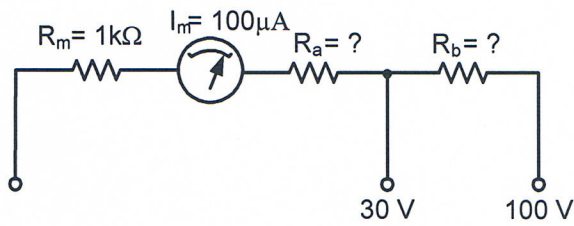
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Q9) An Ammeter range 10A class 2.5, Calculate the maximum %Error for 3 different position of pointer Case (a) = 10A, Case (b) = 5A, Case (c) = 2.5A, (10 marks)

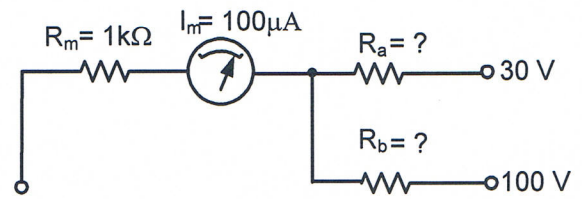


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Q10) A PMMC instrument with $FSD = 100 \mu A$ and $R_m = 1k\Omega$, is to be employed as **a volt-meter** with **ranges of 30 V and 100 V**. **Calculate** the required values of multiplier resistors (R_a and R_b) for both cases and give the **comparison** to the practical volt-meter. (10 marks)



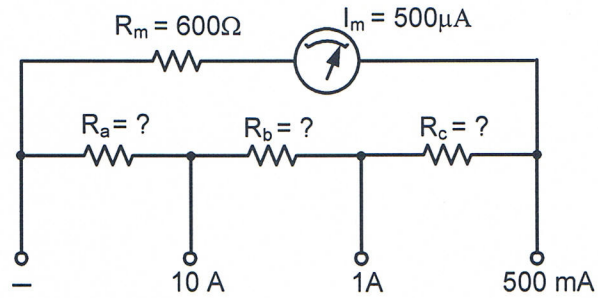
Case (a)



Case (b)

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Q11) A PMMC instrument has $500\ \mu\text{A}$ and a coil resistance of $600\ \Omega$. Calculate the required resistance (R_a , R_b , R_c) values to convert the instrument into **an ammeter** (multi-range) with an FSD of 10A, 1A, and 500mA (10 marks)



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Q12) For designing shunt ohmmeter ($R \times 1$), Calculate expected ohm reading for each scale by using the test meter movement to design an ohmmeter with $R_m = 600 \Omega$, $R_{\text{Series}} = 2.4 \text{ k}\Omega$, $R_{\text{Shunt}} = 20 \Omega$ (10 marks)

- (a) Determine how the resistance scale should be marked at $R_x = 0$ and $R_x = \infty$.
 (b) Determine how the resistance scale should be marked at 0.75 FSD, 0.5 FSD and 0.25 FSD.

